

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester B.Tech. (CBCS) Degree Examinations

December 2022

EE1001-1 – BASIC ELECTRICAL ENGINEERING

Duration: 3 Hours

Max. Marks:100

Note:

1) **Part – A: Multiple Choice Questions:** Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer type Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

2) Assume missing data suitably.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- In the case of mesh analysis, the equations in each loop is written by applying
 A) KVL
 B) KCL
 C) both KCL and KVL
 D) None of these
- While calculating voltage using nodal analysis, it was found that the voltages at nodes V_1 & V_2 were $-5V$ & $-3V$ respectively. Then, the direction of current between those two nodes would be
 A) from V_2 to V_1
 B) from V_1 to V_2
 C) none of these
 D) cannot predict
- Peak factor is defined as ratio of _____ of the alternating quantity.
 A) Maximum value / RMS value
 B) RMS value / Maximum value
 C) RMS value / Average value
 D) Average value / RMS value
- If the instantaneous value of current in a circuit is represented using the equation, $i = 100\sin 120\pi t$ amperes, its RMS value is given by _____
 A) 100 A
 B) $100\sqrt{2}$ A
 C) $100\sqrt{3}$ A
 D) $100/\sqrt{2}$ A
- For a certain load, if the apparent power is 195.2 VA and the reactive power is 125 VAR, then the true power is _____
 A) 70.2 W
 B) 320.2 W
 C) 150W
 D) Data is insufficient
- For a three phase, three wire system, the two Wattmeter read 4000 watts and 2000 watts respectively. Then, the power factor of the circuit is _____
 A) 1
 B) 0.5
 C) 0.866
 D) 0.6
- In a star connected three phase AC circuit _____
 A) $V_{ph} = \sqrt{3}V_L$; $I_L = I_{ph}$
 B) $V_L = \sqrt{3}V_{ph}$; $I_L = I_{ph}$
 C) $V_L = V_{ph}$; $I_L = \sqrt{3}I_{ph}$
 D) $V_L = \sqrt{2}V_{ph}$; $I_L = I_{ph}$
- Which of the following statements is not the definition of power factor (pf)?
 A) $pf = \cos$ of the angle between voltage & current
 B) $pf = \text{resistance} / \text{impedance}$
 C) $pf = \text{active power} / \text{apparent power}$
 D) $pf = \text{apparent power} / \text{active power}$
- The rating of a transformer is specified in _____
 A) kW
 B) kVAR
 C) HP
 D) kVA
- Turns ratio of the transformer is directly proportional to _____
 A) Resistance ratio
 B) power ratio
 C) Voltage ratio
 D) Not proportional to any terms
- The full-load iron loss of a transformer is 3200 W. At 75% of full load, the iron loss will be _____
 A) 3200W
 B) 6400W
 C) 1800W
 D) 5600W
- Transformer core is generally made of _____
 A) silicon steel
 B) aluminium
 C) copper
 D) wood

13. A 4 pole, lap wound, DC generator has a useful flux of 0.07 Wb per pole. Calculate the generated e.m.f. when it is rotated at a speed of 900 r.p.m. with the help of prime mover. Armature consists of 20 slots each having 20 conductors.
 A) 840V B) 420V
 C) 210V D) 21V
14. The number of parallel paths in a 8 pole lap wound DC generator is
 A) 2 B) 16
 C) 8 D) 4
15. A 4 pole, lap wound DC motor drawing an armature current of 20A has 360 conductors. If the flux per pole is 0.015 Wb then the gross torque developed by the armature of motor is
 A) 10.23 Nm B) 15.56Nm
 C) 17.17Nm D) 19.08Nm
16. Alternator works on the following principle
 A) Self and mutual induction B) Mutual induction
 C) Faraday's law of electromagnetic induction D) None of these
17. Which one of the following statements is true?
 A) 3 phase induction motor converts direct current electrical energy into mechanical energy
 B) 3 phase induction motor converts alternating current electrical energy into mechanical energy
 C) 3 phase induction motor converts mechanical energy into alternating current electrical energy
 D) 3 phase induction motor converts mechanical energy into direct current electrical energy
18. The part of the 3 phase induction motor which is a hollow cylindrical core having slots in its inner surface to house windings is termed as:
 A) stator B) rotor
 C) shaft D) brush
19. Fusing factor is defined as the ratio between
 A) maximum fusing current and rated voltage B) maximum fusing current and rated current
 C) minimum fusing current and rated current D) minimum fusing current and rated voltage
20. The objective of earthing or grounding is
 A) to provide as low resistance possible to the ground B) to provide a high resistance possible to the ground
 C) to provide flow of positive, negative and zero sequence current D) none of these

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit - I

Marks BT* CO* PO*

1. a) In the network shown in Fig. 1(a), find the magnitude and direction of current in the various branches by mesh current method.

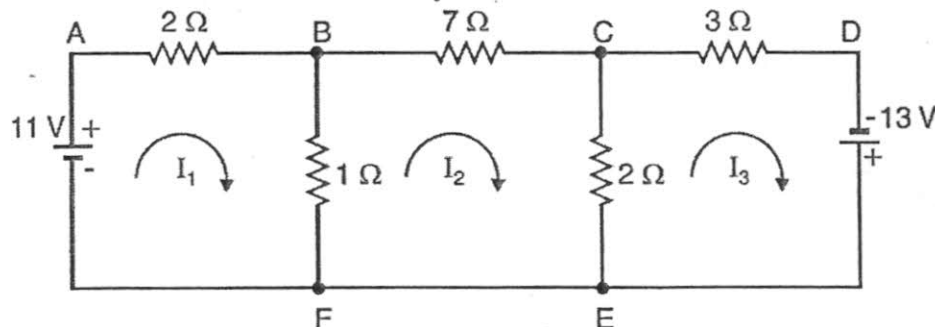


Fig. 1(a)

- b) Explain the relationship between the line and phase quantities in a star connected three phase system with required phasor diagrams. Give the equation for power in the three-phase system in terms of the line quantities.

08 L*3 1 2

08 L2 2 1

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|-------|--|----|----|---|---|
| 2. a) | Define average and RMS value of an alternating quantity. Derive an expression for RMS value considering a sinusoidal alternating voltage in terms of peak value. | 08 | L2 | 1 | 1 |
| b) | A coil having a resistance of 10Ω and an inductance of 35mH is connected to 230V , 50Hz supply. Calculate (i) the impedance and the circuit current (ii) phase angle (iii) power factor (iv) power consumed. | 08 | L3 | 2 | 2 |
| 3. a) | Explain the terms phase and phase difference for sinusoidally varying alternating voltage using suitable demonstrations. | 04 | L2 | 1 | 2 |
| b) | Discuss how an alternating quantity is represented in the phasor form. | 04 | L1 | 1 | 1 |
| c) | Two wattmeters connected to a 3-phase motor indicate the total power input to be 12kW . The power factor is 0.6 . Determine the readings of each wattmeter. | 08 | L3 | 2 | 2 |

Unit – II

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|-------|--|----|----|---|---|
| 4. a) | Define the coefficient of self-inductance and coefficient of mutual inductance. Explain the terms self-induced emf and mutually induced emf. | 05 | L1 | 3 | 1 |
| b) | List the advantages of an autotransformer over two winding transformers. | 03 | L1 | 3 | 1 |
| c) | A 240V , 4 pole shunt motor running at 1000 r.p.m. gives 15HP with an armature current of 50A and field current of 1A . The armature winding is wave connected and has 540 conductors. Its resistance is 0.1Ω and drop at each brush is 1V . Find (i) Useful torque, (ii) Total torque, (iii) Useful flux per pole. | 08 | L3 | 4 | 3 |
| 5. a) | In a 75KVA , single phase transformer the iron and full load copper losses are 500W & 1000W Find (i) Efficiency at 0.8 p.f lagging (ii) The load at which maximum efficiency occurs, (iii) Max efficiency. | 08 | L3 | 3 | 3 |
| b) | Derive the emf equation of an alternator. Give the equation for Frequency of the induced e.m.f. | 08 | L2 | 4 | 1 |
| 6. a) | With a neat sketch, explain the working principle of a single-phase transformer. Derive its emf equation. | 08 | L2 | 3 | 1 |
| b) | A 4-pole lap wound shunt generator supplies to 50 lamps of 100 watts , 200V each. The field and armature resistances are 50Ω and 0.2Ω respectively. Allowing a brush drop of 1V at each brush, calculate the following: (i) Armature current, (ii) Current per path, (iii) Generated e.m.f., (iv) Power output of DC armature. | 08 | L3 | 4 | 2 |

Unit – III

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|-------|---|----|----|---|---|
| 7. a) | Explain the construction and principle of operation of a 3 Phase Induction Motor. | 06 | L2 | 5 | 1 |
| b) | The frequency of the supply applied to 4-pole Induction Motor is 50Hz and that of the rotor induced emf is 1.5Hz . What is the Slip and at what speed the motor is running? | 06 | L3 | 5 | 2 |
| c) | What is an electric fuse? How does it function? | 04 | L1 | 5 | 1 |
| 8. a) | Draw Torque-Slip characteristics of three phase Induction Motor and explain the same. | 06 | L2 | 5 | 2 |
| b) | List the advantages and disadvantages of conduit wiring. | 04 | L1 | 5 | 1 |
| c) | Explain the need of earthing for electrical appliances. With the neat sketch explain the pipe earthing. | 06 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
